Exercise (3.1).

Proof. Let R be a (unital, commutative) ring. Suppose first that every ideal is finitely generated. Then, consider an ascending sequence of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

Then, let

$$I = \bigcup_{n=1}^{\infty} I_n$$

First, note that I is an ideal. Indeed, if $x, y \in I$, then there are n, m with $x \in I_n$ and $y \in I_m$. Then $x, y \in I_{\max\{n,m\}}$, so $x + y \in I_{\max\{n,m\}} \subseteq I$. Further, if $r \in R$ is arbitrary, then $rx \in I_n \subseteq I$. Thus, by assumption, $I = (a_1, \ldots, a_n)$ for some elements $a_i \in R$. Then, for each i, $a_i \in I_{m_i}$ for some indices m_i, and then $a_i \in I_m \subseteq I$ for $m = \max\{m_1, \ldots, m_n\}$. But then $I = I_m$, and so the chain stabilizes: $I = I_m = I_{m+1} = I_{m+2} = \cdots$.

Second, assume that ascending chains stabilize. Let S be a nonempty set of ideals of R. Assume, for contradiction, that S has no maximal element. Then, we can inductively choose a sequence of ideals as follows: let $I_1 \in S$ be arbitrary. Then, since I_1 is not a maximal element of S, there is an ideal $I_2 \in S$ with $I_1 \subsetneq I_2$. Continue in this way: given $I_n \in S$, it is not maximal, so choose $I_{n+1} \in S$ with $I_n \subsetneq I_{n+1}$. But then we've constructed an ascending chain of ideals of R that does not stabilize, contrary to assumption.

Finally, suppose every nonempty collection of ideals has a maximal element. Let $I \subseteq R$ be an ideal, and let $S = \{(a_1, \ldots, a_n) \mid n \in \mathbb{N}, a_i \in I\}$ be the collection of finitely generated ideals contained in I. By assumption, this has a maximal element $(a_1, \ldots, a_n)$. But if $I \neq (a_1, \ldots, a_n)$, then there is some $a \in I \setminus (a_1, \ldots, a_n)$, giving that $(a_1, \ldots, a_n, a) \in S$ strictly contains $(a_1, \ldots, a_n)$. This would contradict maximality, and so we conclude $I = (a_1, \ldots, a_n)$ is finitely generated. □

Exercise (3.2).

Proof. Let R be a domain with $|R|$ finite, and let $\alpha \in R$ be nonzero. Consider the map $\mathbb{N} \to R$ given by $n \mapsto \alpha^n$. Since the codomain is finite, this map cannot be injective. So, there exist distinct $n, m \in \mathbb{N}$ with $\alpha^n = \alpha^m$. WLOG, $n < m$, whence we have

$$0 = \alpha^n(\alpha^{n-m} - 1)$$

But since R is a domain, we either get $\alpha = 0$ or $\alpha^{n-m} = 1$. The former is not true by assumption, and so $\alpha^{n-m} = 1$. So α is invertible and R is a field. □

Exercise (3.3).

Proof. Let G be generated (freely) by $e_1, \ldots, e_n \in G$. Then G/mG is generated by the images $e_i + mG$. Each has order m since $me_i \in mG$, and there are no further relations. For if

$$\sum_{i=1}^{n} a_i(e_i + mG) = mG$$

then $\sum a_i e_i \in mG$, and since these generate freely, each a_i must be a multiple of m. So, the sum is already zero. I.e. G/mG is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^n$ as claimed. □

Exercise (3.4).

Proof. As hinted, if $\alpha \in I$ is nonzero, then $\alpha R \subseteq I \subseteq R$, and so I is an abelian group contained in and containing a free abelian group of rank n. So I is also a free abelian group of rank n. □

Exercise (3.5).

Proof. The two remaining claims in the proof of [Lemma 2, Theorem 15] are that $\gamma \in K \setminus R$ and $\gamma A \subseteq R$. The first is clear, for if $\gamma \in R$, then $b = a\gamma \in (a)$, contrary to assumption. For the second, note that $b \in P_2 \cdots P_r$ and $A \subseteq P = P_1$. Hence $bA \subseteq P_1 \cdots P_r \subseteq aR$. So, $\gamma A \subseteq R$ as claimed. □

Exercise (3.6).

Proof. The only two gaps in the proof are:

- A is an ideal since it is clearly an R-submodule of K contained in R.
- $\gamma J \subseteq J$, since for $\beta \in J$, we have $\gamma\beta \in \gamma J \subseteq R$ and

$$\gamma\beta I \subseteq \gamma I J = \gamma\alpha A \subseteq \alpha R$$

  so that $\gamma\beta \in J$ by definition.

□

Exercise (3.7).

Proof. If $I + J = R$, then there is $a \in I$, $b \in J$ with $a + b = 1$. Then, by the binomial theorem,

$$1 = 1^{n+m} = (a+b)^{n+m} = \sum_{i=0}^{n+m} \binom{n+m}{i} a^i b^{n+m-i}$$

Each summand is either divisible by a^m and so is in I^m or else is divisible by b^n and so is in J^n. So we have $I^m + J^n = R$ as claimed. □

Exercise (3.8).

Proof.

(a) Suppose, for contradiction, that $(2, x) = (f)$ for some $f \in \mathbb{Z}[x]$. Then $2 = fg$ for some $g \in \mathbb{Z}[x]$, so that f has degree zero, i.e. $f \in \mathbb{Z}$. Then $f = \pm 1$ or $f = \pm 2$. We cannot have either of the first two, since then $(f) = \mathbb{Z}[x]$, but

$$\mathbb{Z}[x]/(f) = \mathbb{Z}[x]/(2, x) \cong \mathbb{F}_2$$

is nontrivial. But we also cannot have $f = \pm 2$ because $2 \nmid x$ in $\mathbb{Z}[x]$.

(b) As usual, we refer to the gcd of the coefficients of a polynomial as the content of that polynomial, and refer to any polynomial with content 1 as primitive. As suggested, we first show that the product of primitive polynomials is primitive. Indeed, suppose that $f = \sum_i a_i x^i$ and $g = \sum_j b_j x^j$ are both primitive. Now, let p be a prime, and note that there is some first coefficient of f that is not divisible by p, say a_n and similarly for g, i.e. b_m. But then the coefficient of x^{n+m} in fg is

$$\sum_{i=0}^{n+m} a_i b_{n+m-i}$$

and every term in this sum is divisible by p except the term $a_n b_m$, since it involves a_i for $i < n$ or b_j for $j < m$. So, this coefficient is not divisible by p. I.e. fg is primitive.

Now, for the general case, if m is the content of f and n is the content of g, then 1 is the content of $(f/m)(g/n)$, and so mn is the content of $fg = mn(f/m)(g/n)$.

(c) Contrapositively, suppose $f \in \mathbb{Z}[x]$ is reducible over $\mathbb{Q}$, so that $f = gh$ for nonconstant polynomials $g, h \in \mathbb{Q}[x]$. Then, we can clear denominators: for some $a, b \in \mathbb{Z}$ we get $ag, bh \in \mathbb{Z}[x]$. So,

$$abf = (ag)(bh)$$

i.e. this multiple of f is reducible in $\mathbb{Z}[x]$. Let t be the smallest positive integer such that tf is reducible in $\mathbb{Z}[x]$. If $t \neq 1$, then let p be a prime divisor of t, and note that if $tf = g'h'$, then p divides the content of tf, so it divides the product of the contents of g' and h'. So, it divides one of these, i.e. WLOG p divides the content of g', whence $g'/p \in \mathbb{Z}[x]$. But then $(t/p)f = (g'/p)h'$, contradicting the minimality of t. So we must have $t = 1$ so that $tf = f$ is reducible in $\mathbb{Z}[x]$.

(d) Since f is irreducible in $\mathbb{Z}[x]$, we've shown that f is irreducible in $\mathbb{Q}[x]$. So, $f \mid gh$ implies that $f \mid g$ or $f \mid h$ in $\mathbb{Q}[x]$. WLOG, suppose $f \mid g$, so that $g = fq$ for some $q \in \mathbb{Q}[x]$. As above, by clearing denominators, we can write $ag = f(aq)$ for some $a \in \mathbb{Z}$ such that $aq \in \mathbb{Z}[x]$. Then a divides the content of ag, which equals the content of aq, since f is primitive. So $q = (aq)/a \in \mathbb{Z}[x]$, so that $f \mid g$ in $\mathbb{Z}[x]$.

(e) To see that $\mathbb{Z}[x]$ is a UFD, we will show that every element can be written as a product of irreducibles and that all irreducibles are prime. The former is immediate, since this is true for any Noetherian ring ($\mathbb{Z}[x]$ is Noetherian since $\mathbb{Z}$ is, by an application of the Hilbert Basis Theorem). The latter is essentially what we've shown. Suppose f is irreducible and primitive. Then the above shows that if $f \mid gh$ then $f \mid g$ or $f \mid h$, so that f is prime. If f is not primitive, then $f = d(f/d)$, where d is the content of f. But since f is irreducible, we must have that d is a prime integer and that f/d is a unit, i.e. it is ± 1, so that f itself is $\pm p$ for a prime $p \in \mathbb{Z}$. In this case, $f = p$ is also prime in $\mathbb{Z}[x]$. So, in either case, we've shown that any irreducible is prime, and so $\mathbb{Z}[x]$ is a UFD.

□

Exercise (3.9).

Proof.

(a) Considering the union of all prime divisors of I and J in R, we can write

$$I = P_1^{a_1} \cdots P_n^{a_n} \qquad\qquad J = P_1^{b_1} \cdots P_n^{b_n}$$

for distinct primes P_i and $a_i, b_i \geq 0$. Then, each prime factors in S, so that

$$P_i S = Q_{i1}^{c_{i1}} \cdots Q_{it_i}^{c_{it_i}}$$

where each $c_{iu} > 0$ and as u varies, Q_{iu} enumerates the distinct primes lying over P_i. Further, for $i \neq j$ and any valid u, v, $Q_{iu} \neq Q_{jv}$ since they lie over distinct primes. Overall, this gives:

$$IS = \prod_{i=1}^{n} \prod_{u=1}^{t_i} Q_{iu}^{a_i c_{iu}}$$

and

$$JS = \prod_{i=1}^{n} \prod_{u=1}^{t_i} Q_{iu}^{b_i c_{iu}}$$

Since $IS \mid JS$ and these are all distinct primes, we must have $a_i c_{iu} \leq b_i c_{iu}$ for each i, u. Since each c_{iu} is positive, this gives $a_i \leq b_i$ for each i, and so $I \mid J$.

(b) As suggested, let $J = IS \cap R$. Then, if $x \in I \subseteq R$, then $x \in IS$ also, so $x \in J$, i.e. $I \subseteq J$, so $J \mid I$. But also clearly $JS \subseteq IS$, so $IS \mid JS$, and the previous gives $I \mid J$. Thus $I = J$, i.e. $I = IS \cap R$.

(c) I claim the following is necessary and sufficient for $I = (I \cap R)S$: for each prime P dividing $I \cap R$, there is an n such that for each prime Q of S lying over P, the exponent of Q in the factorization of I is precisely $ne(Q|P)$. Indeed, this is sufficient, for if this holds, then

$$I \cap R = P_1^{n_1} \cdots P_k^{n_k}$$

and then for each prime Q lying over P_i, the exponent of Q in the factorization of $(I \cap R)S$ is $n_i e(Q|P_i)$, which by assumption is the exponent of Q in the factorization of I. Conversely, suppose $I = (I \cap R)S$. Then, if P^n divides $I \cap R$ exactly (i.e. P^{n+1} does not divide $I \cap R$), then writing

$$P = Q_1^{e_1} \cdots Q_t^{e_t}$$

we get that $Q_i^{ne_i}$ exactly divides $(I \cap R)S$, and so the exponent of Q_i in I is exactly $ne_i = ne(Q_i|P)$ as claimed.

□

Exercise (3.10).

Proof. Let $R \subseteq S \subseteq T$ be number rings and let $U \subseteq T$ be prime. Let $Q = S \cap U$ and $P = R \cap Q$ be the corresponding primes lying under. Then QT can be written as a product where one of the factors is $U^{e(U|Q)}$, and PS can similarly be written as a product with one factor equal to $Q^{e(Q|P)}$. Substituting this in gives that $PT = (PS)T$ has a factor of $Q^{e(Q|P)}T = (QT)^{e(Q|P)}$, which has a factor of

$$(U^{e(U|Q)})^{e(Q|P)} = U^{e(U|Q)e(Q|P)}$$

and no higher power, so $e(U|P) = e(U|Q)e(Q|P)$.

Second, note that the inclusions $R \hookrightarrow S \hookrightarrow T$ composed with the quotient $T \to T/U$ induces ring homomorphisms:

$$R/P \to S/Q \to T/U$$

These are nontrivial morphisms and each ring is a field, so they are all injections. I.e. this is a tower of field extensions and the inertial degree is the degree of the extension, so that $f(U|P) = [T/U : R/P] = [T/U : S/Q][S/Q : R/P] = f(U|Q)f(Q|P)$. So, indeed both e, f are multiplicative. $\square$

Exercise (3.11).

Proof. Let $\alpha \in I$. Then $\alpha R \subseteq I$, so $I \mid \alpha R$, so there is some nonzero ideal J with $\alpha R = IJ$. Then

$$\|I\|\|J\| = \|IJ\| = \|\alpha R\| = N^K(\alpha)$$

so that $\|I\| \mid N^K(\alpha)$ as claimed. If these are equal, then $[R : J] = \|J\| = 1$, so $J = R$. Then $\alpha R = IJ = IR = I$ as claimed. Conversely, if $\alpha R = I$, then clearly the two ideals have the same norm. $\square$

Exercise (3.12).

Proof. We show both containments. First, note that

$$(5, \alpha + 2)(5, \alpha^2 + 3\alpha - 1) = (25, 5\alpha + 10, 5\alpha^2 + 15\alpha - 5, 5\alpha^2 + 5\alpha)$$

Clearly this is contained in $5S$ since each generator is. Conversely, this ideal contains

$$(5\alpha + 10)(2\alpha + 2) - 2(5\alpha^2 + 15\alpha - 5) - 25 = 5$$

and so it contains $5S$. So the two are equal.

Consider the map $\mathbb{Z}[x] \to \mathbb{F}_5[x]/(x^2 + 3x - 1)$ (that maps x and 1 to themselves). The kernel clearly contains 5 and $x^2 + 3x - 1$. If f is in the kernel, then by polynomial division we can write $f(x) = q(x)(x^2 + 3x - 1) + r(x)$ for $q, r \in \mathbb{Z}[x]$ and r of degree at most 1 since $x^2 + 3x - 1$ is monic. Then r is in the kernel, and factorizing it (since $\mathbb{Z}[x]$ is a UFD), we find that each prime factor of r is nonzero in the image except possibly 5. Since r is in the kernel, it must be divisible by 5, i.e. $f \in (x^2 + 3x - 1, 5)$, so this is precisely the kernel. I.e. we have an isomorphism:

$$\mathbb{Z}[x]/(5, x^2 + 3x - 1) \cong \mathbb{F}_5[x]/(x^2 + 3x - 1)$$

as claimed.

Consider the map $\mathbb{Z}[x] \to S/(5, \alpha^2 + 3\alpha - 1)$ that maps x to α. Then clearly 5 is in the kernel and $x^2 + 3x - 1 \mapsto \alpha^2 + 3\alpha - 1 = 0$ in the latter ring. So, this factors as a map

$$\mathbb{Z}[x]/(5, x^2 + 3x - 1) \to S/(5, \alpha^2 + 3\alpha - 1)$$

as claimed.

So, $S/(5, \alpha^2 + 3\alpha - 1)$ is (isomorphic to) a quotient of $\mathbb{F}_5[x]/(x^2 + 3x - 1)$. But $x^2 + 3x - 1$ is irreducible in this ring since it has no roots in $\mathbb{F}_5$. So $\mathbb{F}_5[x]/(x^2 + 3x - 1)$ is a field and so the only ideals are the zero and improper ideals, hence the only quotients are itself and the zero ring. So, we get that $(5, \alpha^2 + 3\alpha - 1) = S$ or $S/(\alpha^2 + 3\alpha - 1) \cong \mathbb{F}_{25}$.

Finally, from the first part, if $(5, \alpha^2 + 3\alpha - 1) = S$, then $5S = (5, \alpha + 2)S$. But then $\alpha + 2 \in 5S$, so that $(\alpha + 2)/5 \in S$, which isn't true by unique representation of elements in $\mathbb{Q}[\alpha]$. $\square$

Exercise (3.13).

Proof. As with the previous, we define I and compute:

$$\begin{aligned}
I &= (23, \alpha - 10)^2(23, \alpha - 3) \\
&= (23^2, 23(\alpha - 10), \alpha^2 - 20\alpha + 100)(23, \alpha - 3) \\
&= (23^3, 23^2(\alpha - 10), 23(\alpha^2 - 20\alpha + 100), 23^2(\alpha - 3), 23(\alpha^2 - 13\alpha + 30), \alpha^3 - 23\alpha^2 + 160\alpha - 300) \\
&= (23^3, 23^2(\alpha - 10), 23(\alpha^2 - 20\alpha + 100), 23^2(\alpha - 3), 23(\alpha^2 - 13\alpha + 30), -23\alpha^2 + 161\alpha - 299) \\
&= (23^3, 23^2(\alpha - 10), 23(\alpha^2 - 20\alpha + 100), 23^2(\alpha - 3), 23(\alpha^2 - 13\alpha + 30), -23(\alpha^2 - 7\alpha + 13))
\end{aligned}$$

This is contained in $23S$ since each generator is a multiple of 23. Further, we have:

$$\det \begin{pmatrix} 1 & 1 & 1 \\ -20 & -13 & -7 \\ 100 & 30 & 13 \end{pmatrix} = 7 \cdot 43$$

which is nonzero in $\mathbb{F}_{23}$. So, by linear algebra there are $a, b, c \in \mathbb{Z}$ with

$$a(x^2 - 20x + 100) + b(x^2 - 13x + 30) + c(x^2 - 7x + 13) = 23r(x) + 1$$

for some polynomial $r \in \mathbb{Z}[x]$. Thus, $23^2 r(x) + 23 \in I$. We also have

$$23^2(\alpha - 3) - 23^2(\alpha - 10) = 23^2 \cdot 7 \in I$$

Since 23 and 7 are coprime (in $\mathbb{Z}$), there are $u, v \in \mathbb{Z}$ with $23u + 7v = 1$. Then, $23^3 \in I$, so

$$23^3 u + (23^2 \cdot 7)v = 23^2(23u + 7v) = 23^2 \in I$$

as well. Since $23^2 \in I$ and $23^2 r + 23 \in I$, we finally conclude $23 \in I$, as claimed. Thus finally we get $23S = I$.

We have $(23, \alpha - 10) + (23, \alpha - 3) = (23, \alpha - 10, \alpha - 3)$, so if we can show the latter is S, then they will indeed be coprime ideals. But the latter ideal contains

$$23u + (\alpha - 3)v - (\alpha - 10)v = 23u + 7v = 1$$

and so it is S. □

Exercise (3.14).

Proof. First, note that G acts on the set of primes lying over P. We've shown that this action is transitive; for a prime Q over P, let G_Q denote the stabilizer of Q. Thus, if Q, Q' are two primes lying over P, then there is some $\alpha \in G$ with $\alpha(Q) = Q'$, and so αG_Q is the set of automorphisms that map Q to Q'. But αG_Q and G_Q have the same size, as desired. Thus, if there are r primes lying over P, then $re(Q|P)f(Q|P) = n = r|G_Q|$, so $|G_Q| = e(Q|P)f(Q|P)$ as claimed.

Computing directly, we have, for a fixed Q over P:

$$\begin{aligned} P^{f(Q|P)} &= R \cap P^{f(Q|P)} S \\ &= R \cap \prod_{\substack{Q' \in \operatorname{Spec} S \\ Q \supseteq P}} (Q')^{e(Q'|P)f(Q|P)} \\ &= R \cap \prod_{\sigma \in G} \sigma(Q) \\ &= N_K^L(Q) \end{aligned}$$

since $f(Q|P) = f(Q'|P)$ for any Q' lying over P.

Note that if σ is an automorphism of L/K and I, J are ideals of S, then $\sigma(IJ) = \sigma(I)\sigma(J)$. So, let I be a nonzero ideal of S, so we can factorize it as $I = \prod_{i=1}^n Q_i$, where each Q_i is a prime lying over $P_i = Q_i \cap R$. So,

$$\prod_{\sigma \in G} \sigma(I) = \prod_{\sigma \in G} \prod_{i=1}^n \sigma(Q_i) = \prod_{i=1}^n P_i^{f(Q_i|P_i)} S$$

so the product is JS for $J = \prod_{i=1}^n P_i^{f(Q_i|P_i)}$. Then,

$$J = R \cap JS = R \cap \prod_{\sigma \in G} \sigma(I) = N_K^L(I)$$

and so

$$\prod_{\sigma \in G} \sigma(I) = JS = N_K^L(I)S$$

as claimed.

Directly, we have:

$$\begin{aligned}
N_K^L(IJ) &= R \cap N_K^L(IJ)S \\
&= R \cap \prod_{\sigma \in G} \sigma(IJ) \\
&= R \cap \prod_{\sigma \in G} \sigma(I)\sigma(J) \\
&= R \cap \left[\left(\prod_{\sigma \in G} \sigma(I) \right) \left(\prod_{\sigma \in G} \sigma(J) \right) \right] \\
&= \left[\left(R \cap \prod_{\sigma \in G} \sigma(I) \right) \left(R \cap \prod_{\sigma \in G} \sigma(J) \right) \right] \\
&= N_K^L(I) N_K^L(J)
\end{aligned}$$

as desired.

Again, we do this directly:

$$N_K^L(\alpha S) = R \cap \prod_{\sigma \in G} \sigma(\alpha S) = R \cap \left(\prod_{\sigma \in G} \sigma(\alpha) \right) S = R \cap N_K^L(\alpha) S = N_K^L(\alpha) R$$

as claimed. □

Exercise (3.15).

Proof. Note that it suffices to show the claim for primes, since both sides are multiplicative. So, let $R \subseteq S \subseteq T$ be the rings of integers in K, L, M, respectively, let $U \in \operatorname{Spec} T$, let $Q = U \cap S$, and let $P = Q \cap R$. Then,

$$N_K^L(N_L^M(U)) = N_K^L(Q^{f(U|Q)}) = N_K^L(Q)^{f(U|Q)} = (P^{f(Q|P)})^{f(U|Q)} = P^{f(U|Q)f(Q|P)} = P^{f(U|P)} = N_K^M(U)$$

as claimed.

As suggested, now let M be the normal closure of L/K. Then for $\alpha \in S$, and $n = [M : L]$, $N_L^M(\alpha) = \alpha^n$, so

$$N_K^L(\alpha^n S) = N_K^L(N_L^M(\alpha T)) = N_K^M(\alpha T) = N_K^M(\alpha) R = [N_K^L(\alpha)]^n R$$

using the result of the previous problem and the fact that M/L and M/K are normal. Since the norm is multiplicative and prime factorizations are unique, $N_K^L(\alpha S) = N_K^L(\alpha) R$ as claimed.

As suggested, note that if $P \in \operatorname{Spec} S$, then P lies over a rational prime $p \in \mathbb{Z}$, and

$$N_{\mathbb{Q}}^L(P) = (p\mathbb{Z})^{f(P|p)} = p^{f(P|p)}\mathbb{Z} = |S/P|\mathbb{Z} = \|P\|\mathbb{Z}$$

as claimed. Further, both sides are multiplicative, so this is now true for all ideals I of S. □

Exercise (3.16).

Proof. We have a map of ideals $I \mapsto N_K^L(I)$. To show that it induces a map of class groups, we show that it maps equivalent elements to equivalent elements. So, let A, B be nonzero ideals of S with nonzero elements $a, b \in S$ with $aA = bB$. Then,

$$N_K^L(a) N_K^L(A) = N_K^L(aA) = N_K^L(bB) = N_K^L(b) N_K^L(B)$$

and so $N_K^L(A)$ and $N_K^L(B)$ are equivalent.

For an ideal I, let $[I]$ denote the class of I in the class group. Then:

$$[R] = N_K^L([S]) = N_K^L([Q]^{d_Q}) = [N_K^L(Q)^{d_Q}] = [P^{f(Q|P)d_Q}]$$

so that $d_P \mid f(Q|P)d_Q$ as claimed. □

Exercise (3.17).

Proof. Throughout, let $S = \mathbb{Z}[\omega]$.

First, we have that $f(Q|2)$ is the multiplicative order of 2 mod 23, which is 11. Then, $f(Q|P)f(P|2) = f(Q|2) = 11$, and $f(P|2) \leq [K : \mathbb{Q}] = 2$, so we must have $f(P|2) = 1$ and $f(Q|P) = 11$. Then, since $[L : K] = [L : \mathbb{Q}]/[K : \mathbb{Q}] = 22/2 = 11$, we get that $PS = Q$, since the sum of $e \cdot f$ over all primes lying over P gives 11. I.e. $Q = (2R + \theta R)S = 2S + \theta S$ as claimed.

We have:

$$P^3 = (8, 4\theta, 2\theta^2, \theta^3) = (8, 4\theta, 2\theta - 12, 5\theta + 6)$$

since $\theta^2 = \theta - 6$. This equation also gives:

$$(\theta - 2)^2 = \theta^2 - 4\theta + 4 = -3\theta - 2 = -3(\theta - 2) - 8$$

So, $\theta - 2$ divides 8. Then it also divides $4(\theta - 2) + 8 = 4\theta$, $2(\theta - 2) - 8 = 2\theta - 12$, and $5(\theta - 2) + 16 = 5\theta + 6$. Thus $P^3 \subseteq (\theta - 2)$.

Conversely, $\theta - 2 = (5\theta + 6) - (4\theta) - (8)$, so $\theta - 2 \in P^3$. Thus, the two ideals are equal, as claimed.

On the other hand, suppose that P is principal, generated by α. Then $P^3 = (\alpha^3) = (\theta - 2)$, and so

$$8 = |N_{\mathbb{Q}}^K(\theta - 2)| = \|P\| = |N_{\mathbb{Q}}^K(\alpha^3)| = |N_{\mathbb{Q}}^K(\alpha)|^3$$

and so $N_{\mathbb{Q}}^K(\alpha) = \pm 2$. But if $\alpha = a + b\theta$, then its norm is:

$$(a + b\theta)(a + b(1 - \theta)) = a^2 + ab + b^2(\theta - \theta^2) = a^2 + ab + 6b^2 = \frac{a^2 + 11b^2 + (a + b)^2}{2}$$

So, in order for this to be ± 2, we would need $a^2 + 11b^2 + (a + b)^2 = \pm 4$, which must actually be 4. This forces $b = 0$, else it would be too large, and so $2a^2 = 4$, which has no integer solutions. So, P is indeed not principal.

From the previous exercise, $3 = d_P \mid d_Q f(Q|P) = 11 d_Q$. So, $3 \mid d_Q$ and Q is not principal.

Suppose $2 = \alpha\beta$ for some nonunits $\alpha, \beta \in S$. Then, $(\alpha)(\beta) = 2S = QQ'$, where Q is as above and $Q' = (2, 1 - \theta)$ lies over the other prime $P' = (2, 1 - \theta$ over 2. But then comparing prime factorizations gives that (WLOG) $Q = (\alpha)$, contradicting the fact that Q is not principal. $\square$

Exercise (3.18).

Proof. Let $\sigma_1, \ldots, \sigma_n$ be the embeddings of K into $\mathbb{C}$. Then:

$$\begin{aligned}
\operatorname{disc}(r\alpha_1, \alpha_2, \ldots, \alpha_n) &= \det \begin{pmatrix} r\sigma_1(\alpha_1) & r\sigma_2(\alpha_1) & \cdots & r\sigma_n(\alpha_1) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \cdots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \sigma_2(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 \\
&= r^2 \det \begin{pmatrix} \sigma_1(\alpha_1) & \sigma_2(\alpha_1) & \cdots & \sigma_n(\alpha_1) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \cdots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \sigma_2(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 \\
&= r^2 \operatorname{disc}(\alpha_1, \ldots, \alpha_n)
\end{aligned}$$

as claimed.

Similarly, if $\beta = c_2\alpha_2 + \cdots + c_n\alpha_n$:

$$\begin{aligned}
\operatorname{disc}(\alpha_1+\beta, \alpha_2, \ldots, \alpha_n) &= \det \begin{pmatrix} \sigma_1(\alpha_1+\beta) & \sigma_2(\alpha_1+\beta) & \cdots & \sigma_n(\alpha_1+\beta) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \cdots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \sigma_2(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{pmatrix}^2 \\
&= \left[\det \begin{pmatrix} \sigma_1(\alpha_1) & \sigma_2(\alpha_1) & \cdots & \sigma_n(\alpha_1) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \cdots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \sigma_2(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{pmatrix} + \sum_{j=2}^{n} c_j \det \begin{pmatrix} \sigma_1(\alpha_j) & \sigma_2(\alpha_j) & \cdots & \sigma_n(\alpha_j) \\ \sigma_1(\alpha_2) & \sigma_2(\alpha_2) & \cdots & \sigma_n(\alpha_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_1(\alpha_n) & \sigma_2(\alpha_n) & \cdots & \sigma_n(\alpha_n) \end{pmatrix} \right]^2 \\
&= \operatorname{disc}(\alpha_1, \ldots, \alpha_n)
\end{aligned}$$

since each term of the sum is zero, as it contains a repeated row.

□

Exercise (3.19).

Proof. As suggested, let $\overline{\alpha} \in R/P$ and $\overline{\beta} \in S/PS$ be the images of α, β under the quotient maps. Since S/PS is an R/P-vector space, the equation $\overline{\alpha}\overline{\beta} = 0$ implies that $\overline{\alpha} = 0$, so $\alpha \in P$, or else $\overline{\beta} = 0$, so $\beta \in PS$.

More directly, if $\alpha\beta \in PS$ and we assume $\beta \notin P$, then since P is maximal, $R\beta + P = R$, so we can find $r \in R$ and $p \in P$ with $r\beta + p = 1$. Then,

$$\alpha = r\alpha\beta + p\alpha \in PS$$

since both summands are.

Note that if $\beta_i \notin P$ for some i, then $\gamma = 1$ works trivially. So, we may assume $\beta_i \in P$ for each i, so that $\alpha\beta \in PS$. But $\alpha \notin PS$, so the previous argument gives $\beta \in P$ as well. Thus, $B = (\beta, \beta_1, \ldots, \beta_n) \subseteq P$. By a lemma from the chapter, there is some $\gamma \in K$ with $B\gamma \subseteq R$ but $B\gamma \not\subseteq P$. It is clear that $\beta\gamma \in R$ and $\beta_i\gamma \in R$ for each i. So, we need only show that it isn't the case that $\beta_i\gamma \in P$ for all P. Suppose this is the case; then

$$\alpha(\beta\gamma) = \sum_{i=1}^{n} \alpha_i(\beta_i\gamma) \in PS$$

and so the previous result again gives $\beta\gamma \in P$. But then $B\gamma \subseteq P$, contrary to assumption. So, $\beta_i\gamma \notin P$ for some i.

For the claim, we imitate the proof of theorem 24. Since P is ramified in S, the factorization of PS contains a repeated prime. Removing that prime, we get an ideal I of S such that $I \supsetneq PS$ and such that each prime lying over P divides I. Now, I contains PS properly, so choose $\alpha \in I \setminus PS$, and using the fact that the α_i form a basis, write:

$$\alpha = c_1\alpha_1 + \cdots + c_n\alpha_n$$

for some $c_i \in K$. Clearing denominators gives:

$$\alpha\beta = \alpha_1\beta_1 + \cdots + \alpha_n\beta_n$$

for $\beta, \beta_i \in R$ and $c_i = \beta_i/\beta$. By the previous, after multiplying by an element $\gamma \in K$ if necessary, we may assume that not all $\beta_i \in P$, and after rearranging we may assume $\beta_1 \notin P$. Then, from a previous exercise, we have:

$$\operatorname{disc}_K^L(\alpha, \alpha_2, \ldots, \alpha_n) = \operatorname{disc}_K^L(\alpha_1\beta_1, \alpha_2, \ldots, \alpha_n) = \beta_1^2 \operatorname{disc}_K^L(\alpha_1, \ldots, \alpha_n)$$

So, to show that $\operatorname{disc}_K^L(\alpha_1, \ldots, \alpha_n) \in P$, it suffices to show that $d = \operatorname{disc}_K^L(\alpha, \alpha_2, \ldots, \alpha_n) \in P$, since P is prime and $\beta_1 \notin P$.

Let M be a normal extension of L/K, fix a prime Q of (the ring of integers of) M, and let σ be a K-embedding of L into $\mathbb{C}$. Then σ extends to an automorphism of M, and so $\sigma^{-1}(Q)$ is also a prime lying over P. So $\sigma^{-1}(Q) \cap S$ is a prime of S lying over P, which divides (and thus contains) I, and so $\alpha \in \sigma^{-1}(Q)$. This shows that $\sigma(\alpha) \in Q$ for each σ, and so expanding d as a determinant shows that $d \in Q$ as well. But we also have that $d \in R$, and so $d \in R \cap Q = P$ as desired. □

Exercise (3.20).

Proof. Let $f_i = f(Q_i|P)$, and write $B_i = \{\beta_{i1}, \ldots, \beta_{if_i}\}$. Let

$$s = \sum_{i=1}^{r} \sum_{j=1}^{e_i} \sum_{k=1}^{f_i} c_{ijk}\alpha_{ij}\beta_{ik}$$

for some $c_{ijk} \in R$, and suppose $s \in P$. We'd like to show that each $c_{ijk} \in P$. Consider this equation mod Q_i. Then since $s \in P \subseteq PS \subseteq Q_i$:

$$0 \equiv s \equiv \sum_{k=1}^{f_i} c_{i1k}\alpha_{i1}\beta_{ik} = \alpha_{i1} \sum_{k=1}^{f_i} c_{i1k}\beta_{ik} \pmod{Q_i}$$

since $\alpha_{hj} \in Q_i$ for $h \neq i$ as well as for $h = i$ and $j > 1$. But $\alpha_{i1} \neq 0$ since it isn't in Q_i, so the sum must be zero. But β_{ik} is a basis for S/Q_i over R/P (as k varies), so we conclude $c_{i1k} \in P$ for all k. Since i was arbitrary, $c_{i1k} \in P$ for all i, k.

Now, suppose we've shown $c_{irk} \in P$ for all r less than some $j > 1$. Then we'll show $c_{ijk} \in P$ for all i, k. For this, consider s modulo Q_i^j. Then, again $s \in P \subseteq PS \subseteq Q_i^{e_i} \subseteq Q_i^j$, so:

$$0 \equiv s \equiv \sum_{k=1}^{f_i} c_{ijk}\alpha_{ij}\beta_{ik} = \alpha_{ij} \sum_{k=1} c_{ijk}\beta_{ik} \pmod{Q_i^j}$$

Now, $\alpha_{ij} \notin Q_i^j$, so we must have that the sum is in Q_i. Again using the fact that β_{ik} forms a basis gives $c_{ijk} \in P$ for all i, k. By induction $c_{ijk} \in P$ for all i, j, k, as claimed. □

Exercise (3.21).

Proof. As suggested, suppose $p \mid |S/G|$, so there is some $a \in S \setminus G$ such that $pa \in G$. I.e. we can write:

$$pa = c_1\alpha_1 + \cdots + c_n\alpha_n$$

for some $c_i \in \mathbb{Z}$. But since the α_i are independent, we conclude that each $c_i \in p\mathbb{Z}$, i.e. $c_i = pd_i$ for some $d_i \in \mathbb{Z}$. Then

$$a = d_1\alpha_1 + \cdots + d_n\alpha_n \in G$$

contrary to assumption.

Then,

$$\operatorname{disc}(\alpha_1, \ldots, \alpha_n) = \operatorname{disc}(G) = |S/G|^2 \operatorname{disc}(S)$$

which is the claim, since $m = |S/G|^2$ is not divisible by p.

Now, let M be a normal extension of $L/\mathbb{Q}$ and let T be its ring of integers. Extend the K-embeddings of L into $\mathbb{C}$ to automorphisms $\sigma_1, \ldots, \sigma_n$ of M. From the previous problem, we are considering the elements $\alpha_{ij}\beta_{ik}$. Then $\operatorname{disc}(\alpha_{ij}\beta_{ik}) = \det(A)^2$, where A is the matrix consisting of elements $\sigma_t(\alpha_{ij}\beta_{ik})$. Fix a prime U of T lying over Q_1 and let $e = e(U|P)$ be the common ramification index; we'd like to compute $v_U(\det(A))$, the power of U appearing in the factorization of $\det(A)T$. Since each automorphism permutes the primes in T, $v_U(\sigma(x))$ is at least as large as the smallest exponent occuring in the factorization of xT. Thus:

$$v_U(\sigma_t(\alpha_{ij}\beta_{ik})) \geq v_U(\sigma_t(\alpha_{ij})) \geq \min\left(\left\{\frac{e}{e_i}(j-1)\right\} \cup \left\{\frac{e}{e_h}N \mid h \neq i\right\}\right) = \frac{e}{e_i}(j-1)$$

as long as N is large enough.

Thus, when computing the valuation of $\det(A)$, we can factor out at least $U^{e(j-1)/e_i}$ from the column corresponding to $\alpha_{ij}\beta_{ik}$. Thus, we get:

$$v_U(\det(A)) \geq \sum_{i=1}^{r} \sum_{j=1}^{e_i} \sum_{k=1}^{f_i} \frac{e(j-1)}{e_i} = \frac{e}{2} \sum_{i=1}^{r} f_i(e_i - 1)$$

and so

$$v_p(\det(A)^2) = \frac{2}{e} v_U(\det(A)) \geq \sum_{i=1}^{r} f_i(e_i - 1) = k$$

as claimed. The first part of the problem gives that $p^k \mid \operatorname{disc}(S)$ as well, since $\det(A)^2/\operatorname{disc}(S)$ is a p-free integer. □

Exercise (3.22).

Proof. Suppose $\alpha^5 - 2\alpha - 2 = 0$. Then

$$\operatorname{disc}(\alpha) = 5^5(-2)^4 + 4^4(-2)^5 = 2^4 \cdot 3 \cdot 13 \cdot 67$$

Let $R = \mathbb{Z}[\alpha]$ and S be the ring of integers in $\mathbb{Q}(\alpha)$. We know $R \subseteq S$ and

$$\operatorname{disc}(R) = \operatorname{disc}(S)|S/R|^2$$

So, we have that $|S/R|$ is a power of two, since the square of no other prime divides $\operatorname{disc}(R)$. So, finally, we seek the power of 2 dividing $\operatorname{disc}(S)$. The previous problem suggests studying the factorization of $2S$. But from 2.43, $\alpha + 1$ is a unit, and so

$$(\alpha S)^5 = \alpha^5 S = (2\alpha + 2)S = 2(\alpha + 1)S = 2S$$

So, the factorization of $2S$ is given by the factorization of αS raised to the fifth. But the exponents in the factorization of $2S$ sum to at most $n = 5$, so this must be the factorization itself. I.e. αS is the unique prime lying over 2 with $e = 5$ and $f = 1$. By the previous exercise, $\operatorname{disc}(S)$ is divisible by $2^{(5-1)\cdot 1} = 2^4$.

This completes the computation, showing that $|S/R| = 1$, i.e. $S = R = \mathbb{Z}[\alpha]$.

Now consider the case $\alpha^5 + 2\alpha^4 - 2 = 0$; we consider 2.44 now. Then,

$$\operatorname{disc}(\alpha) = -2^4 \cdot 971$$

So, as before, we consider $2S$. But

$$(\alpha S)^5 = \alpha^5 S = (-2\alpha^4 + 2)S = 2(\alpha^4 - 1)S = 2S$$

since $\alpha^4 - 1$ is a unit. So, this is the factorization, and the previous problem gives that if S is the ring of integers in $\mathbb{Q}(\alpha)$, then $\operatorname{disc}(S)$ is divisible by 2^4 as well. So,

$$|S/\mathbb{Z}[\alpha]|^2 = \operatorname{disc}(\alpha)/\operatorname{disc}(S) \mid 971$$

so that $S = \mathbb{Z}[\alpha]$ as before. □

Exercise (3.23).

Proof. We establish each of the missing parts. First, 3.2. We have:

$$(2, 1 + \sqrt{m})^2 = (4, 2 + 2\sqrt{m}, m + 1 + 2\sqrt{m})$$

This is contained in $2R$ since m is odd. Since $m \equiv 3 \pmod 4$, we have $m = 3 + 4k$ for some k, so this ideal also contains:

$$(m + 1 + 2\sqrt{m}) - (2 + 2\sqrt{m}) - k(4) = m - 1 - 4k = 2$$

So, it contains $2R$ and the two are equal.

For 3.3, we have $m = 8k + 1$ for some k, so:

$$\left(2, \frac{1 + \sqrt{m}}{2}\right)\left(2, \frac{1 - \sqrt{m}}{2}\right) = \left(4, 1 + \sqrt{m}, 1 - \sqrt{m}, \frac{1 - m}{4}\right) = (4, 1 + \sqrt{m}, 1 - \sqrt{m}, -2k)$$

This is again clearly contained in $2R$ (since $(1 \pm \sqrt{m})/2 \in R$). But it also contains $(1 + \sqrt{m}) + (1 - \sqrt{m}) = 2$, so it contains $2R$.

Finally, for 3.5, we have $n^2 = m + kp$ for some k, so

$$(p, n + \sqrt{m})(p, n - \sqrt{m}) = (p^2, p(n + \sqrt{m}), p(n - \sqrt{m}), kp)$$

which is clearly contained in pR. Conversely, it contains

$$p(n + \sqrt{m}) + p(n - \sqrt{m}) = 2np$$

and so it contains $\gcd(2np, p^2) = p$, since $p \nmid 2n$ as p is an odd prime and $p \nmid m$. So it contains pR. □

Exercise (3.24).

Proof. □

Exercise (3.25).

Proof. □

Exercise (3.26).

Proof. □

Exercise (3.27).

Proof. □

Exercise (3.28).

Proof. □

Exercise (3.29).

Proof. □

Exercise (3.30).

Proof. □

Exercise (3.31).

Proof. □

Exercise (3.32).

Proof. □

Exercise (3.33).

Proof. □

Exercise (3.34).

Proof. □

Exercise (3.35).

Proof. □

Exercise (3.36).

Proof. □

Exercise (3.37).

Proof. □

Exercise (3.38).

Proof. □

Exercise (3.39).

Proof. □

Exercise (3.40).

Proof. □